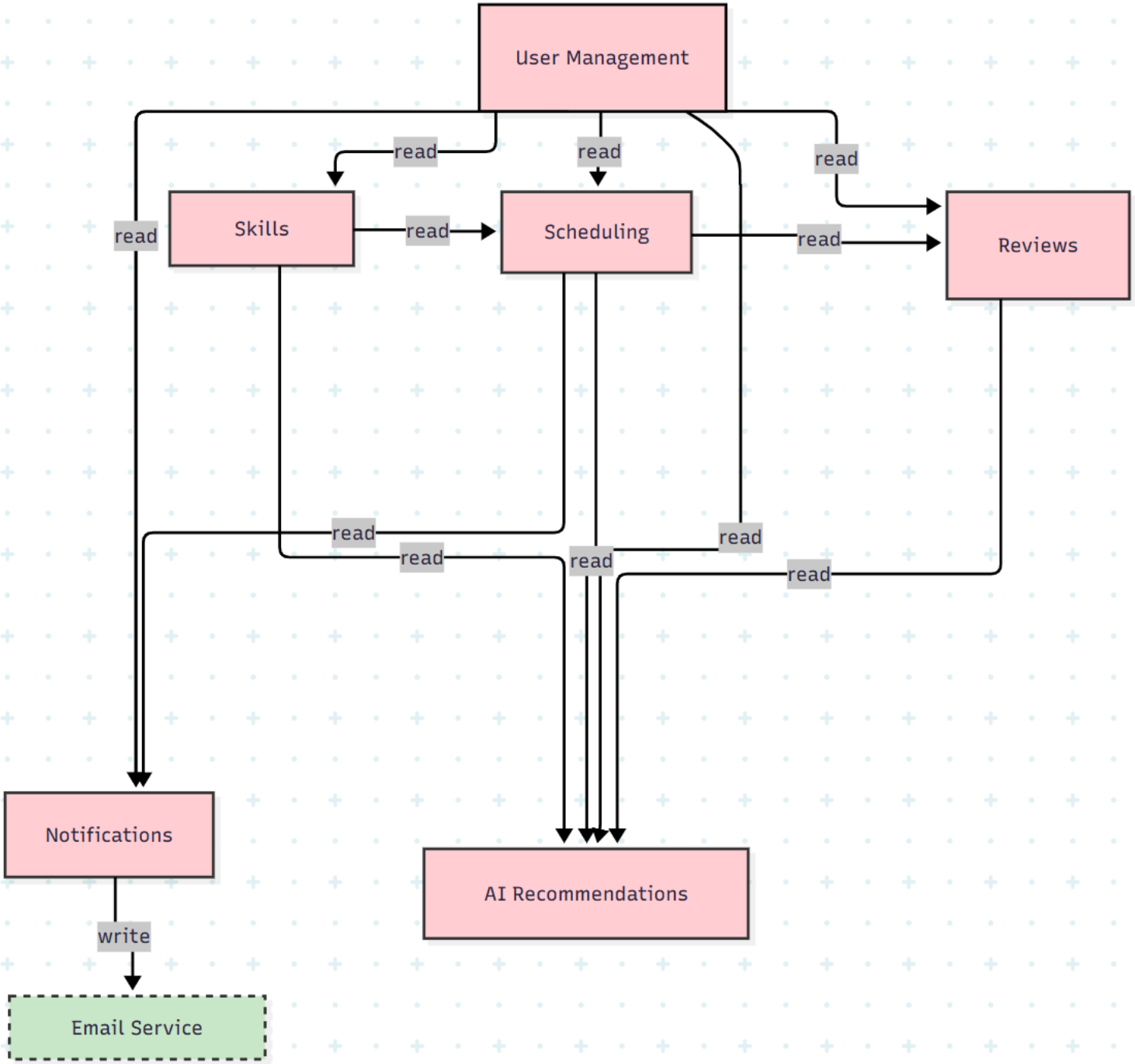
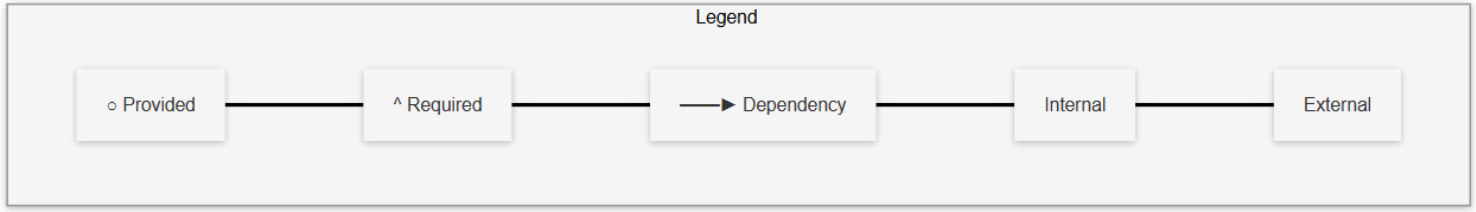
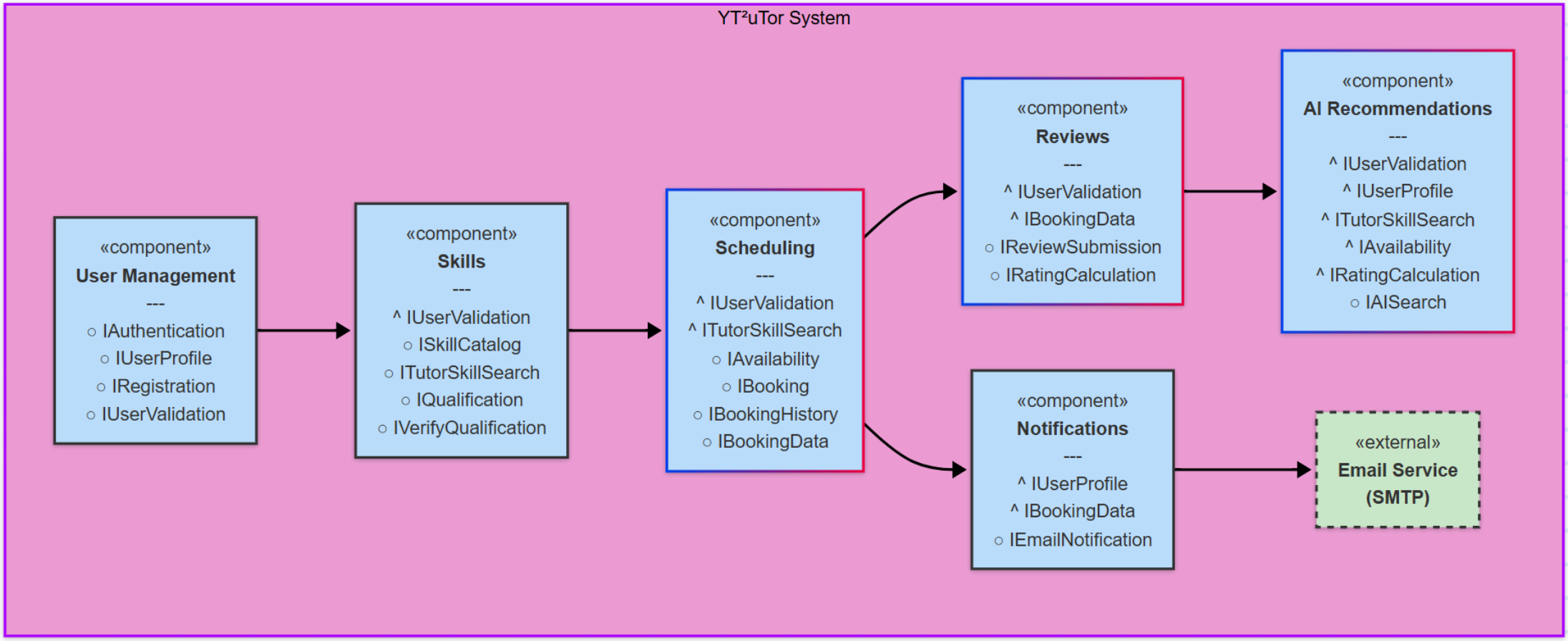
YT²uTor – System Decomposition

Subsystem	Responsibility	Key Use Cases / Features	Depends On	Justification
User Management	Manage student and tutor accounts, registration, login, profiles, and authentication	Register (Student/Tutor), Login, Manage Profile, View Profile, Authentication	— (core entity owner)	Every system needs user management. It owns user data and authentication. No other component should directly modify user credentials or roles.
Skills	Manage the skill catalog, tutor skill associations, qualification documents (certificates), admin verification of tutor qualifications, and enable students to search for tutors by skill	Add Skill, Remove Skill, Search by Skill, Upload Qualification, Verify Qualification (Admin), View Qualifications	User Management	Qualifications exist to prove skills. A tutor's skills and their proof belong together in business logic. Splitting them would require cross-component coordination for every skill addition. Search by skill is a query on this same data – not a separate component.
Scheduling	Manage tutor availability time slots, booking requests, confirmations, rejections, cancellations, booking history, and email notifications for booking events	Set Availability, View Availability, Send Booking Request, View Booking Requests, Confirm/Reject Booking, Cancel Booking, View Bookings (Student), View Schedule (Tutor), View Booking History	User Management, Skills	Availability and bookings operate on the same entities (time slots). A booking cannot exist without an availability slot. Email notifications are triggered by booking events – not complex enough for a separate component.
Notifications	Manage all email communications triggered by the system, including booking confirmations, booking rejections, and password reset emails	Send Booking Confirmation Email, Send Booking Rejection Email, Send Password Reset Email	User Management (for password reset), Scheduling (for booking emails)	Email logic is shared across multiple subsystems (Scheduling and User Management). Centralizing it avoids duplication and makes it easier to change email behavior without modifying multiple components.
Reviews	Manage student ratings, comments, and average rating calculation for tutors	Leave Review, View Reviews, Calculate Average Rating	User Management, Scheduling	Reviews have an independent lifecycle. They only depend on completed bookings, not on active bookings or availability. They can be developed separately and don't affect core booking flow.
AI Recommendations	Process natural language student requests, extract skills, detect urgency (keywords like "exam", "tomorrow"), and return top 3 tutor suggestions	Natural Language Search, Urgency Detection, Top 3 Tutor Suggestions	User Management, Skills, Scheduling, Reviews	Regular search returns all matching tutors. AI search applies additional business rules: urgency detection (exam tomorrow → prioritize available tutors), ranking by rating, and limiting to top 3. This is a different business decision process.

Subsystem Dependency Map



High Level Component Diagram